

Markscheme

Mathematics: analysis and approaches
Practice question bank

113. (a)

Given that $(1 - 2x)^{-\frac{1}{2}} \approx 1 + x + \frac{3}{2}x^2$ for small values of x .

Using $x = 0.01$, estimate $\frac{1}{\sqrt{0.98}}$ correct to 4 decimal places.

note $1 - 2(0.01) = 0.98$ *(M1)*

substitute into the approximation: $1 + 0.01 + 1.5(0.01)^2$ *(M1)*

$= 1.01015$ **A1**

≈ 1.0102 **A1**

[5 marks]